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and the farmer crosses the river with goat. Model the action with an assignment statement.

4. Specify a function to find the minimum of two numbers.

5. If $\sqrt{2}$ = 1.414, and the square_root() function returns -1.414, does it violate the following specification?

   -- square_root (x)

   -- inputs: x is a real number , x ≥ 0

   -- outputs: y is a real number such that y^2=x

Part III Short Answers

1. When do you say that a problem is algorithmic in nature?
2. What is the format of the specification of an algorithm?
3. What is abstraction?
4. How is state represented in algorithms?
5. What is the form and meaning of assignment statement?
6. What is the difference between assignment operator and equality operator?

Part IV Explain

1. Write the specification of an algorithm hypotenuse whose inputs are the lengths of the two shorter sides of a right angled triangle, and the output is the length of the third side.
2. Suppose you want to solve the quadratic equation ax2 + bx + c = 0 by an algorithm.

   quadratic_solve (a, b, c)

   -- inputs : ?

   -- outputs: ?

   You intend to use the formula and you are prepared to handle only real number roots. Write a suitable specification.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$